

AGNI-NETRA

AI Geospatial Network for Industrial Thermal Risk & Anomaly Analysis

INTELLIGENCE DOSSIER • Generated on 2026-09-14 13:32 UTC • DECISION SUPPORT SYSTEM

Event Code: EVT-GUJ-20260830-3112	Classification: Industrial Fire (97.1%)
Location: 22.35420°N, 69.86440°E (Gujarat)	AGNI-NETRA Risk Level: CRITICAL (80.3/100)
Active Detections: 6 observations	Peak Radiative Power (FRP): 285.0 MW
First Detected: 2026-08-30 04:09:35	Last Detected: 2026-08-30 04:09:35

1. Explainable AI (SHAP) Attribution

Evaluated via candidate XGBoost v3.0 model with TreeExplainer SHAP attribution.

Feature Name	Value	Shapley Impact	Direction
dist_to_facility_m	181.91402943653523	+0.420	Supports Class (+)

2. Geospatial Facility Association

Associated Facility: None (Uncataloged Site)	Facility Type: N/A
Facility Status: KNOWN	Distance to Boundary: 181.9 m
Land Cover Context: Industrial	Source Provenance: OSM / Satellite Survey

3. Risk Evaluation & Drivers

- Severe radiative heat output (Peak FRP: 285.0 MW)
- Current FRP (285.0 MW) is statistically critical (+6.4σ above historical baseline mean of 125.0 MW).
- Chronic persistent thermal source with continuous multiday emissions
- Direct correlation with high-hazard industrial facility boundary

4. Model Governance & Data Provenance

Model Version: XGBoost V3 (xgb-v3.0-real-candidate • Platt-Calibrated)	Telemetry Source: NASA FIRMS S-NPP/VIIRS 375m NRT
Spatial Context Engine: PostGIS 3.4 Multi-Layer Topology	Verification Status: ACTIVE (Pending Human Verification)
Automated Dispatch: DISABLED / GATED SAFE (Human Decision Support)	Dossier Generated: 2026-09-14 13:32:18 UTC

LEGAL & SCIENTIFIC DISCLAIMER: This document contains automated analytical intelligence derived from satellite remote sensing observations and machine learning models. Predictions reflect statistical probabilities and do not constitute certified ground truth until verified by authorized domain personnel.